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## \* Simple linear regression (SLR)

$$(X_i, Y_i), i = 1, 2, \dots, n$$

Q) What does linear regression model?

A) Given  $X=x$ , what's avg. value of  $Y$ :

$$E(Y|X=x) = \text{linear func}^n \text{ of } x = \beta_0 + \beta_1 x$$

• A regression model involves,

a) a description of the conditional mean of  $Y$  given  $X$ .

b) " " " " " variability " " " "

c) " " " " " distribution " " " "

• Methods for fitting SLR:

$$\textcircled{1} \text{ Least squares: } \sum (Y_i - \beta_0 - \beta_1 X_i)^2$$

$$\textcircled{2} \text{ Least absolute deviation: } \sum |Y_i - \beta_0 - \beta_1 X_i|$$

$$\textcircled{3} \text{ Power loss: } \sum |Y_i - \beta_0 - \beta_1 X_i|^\alpha, \alpha \geq 1$$

## \* Poisson Regression Model (PRM)

$$Y|X=x \sim \text{Poisson}(\beta_0 + \beta_1 x)$$

$$E(Y|X=x) = V(Y|X=x) = \beta_0 + \beta_1 x$$

• Least squares estimator: (through partial derivatives)

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

$$\hat{\beta}_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2} = \frac{S_{xy}}{S_x^2} = r_{xy} \cdot \frac{S_y}{S_x}$$

• A diff. way to look at  $\hat{\beta}_1$ :

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X}) \cdot Y_i}{\sum_{j=1}^n (X_j - \bar{X})^2}$$

A weighted sum of  $Y_i$ 's.